

A school-oriented explanation of teachers' retention decisions: Building on evidence from a systematic review of reviews

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Abstract

Teacher retention remains a pressing challenge in many education systems worldwide despite the many longstanding policy and research initiatives aimed at addressing the challenge. The study had three objectives. The first objective was to provide a comprehensive summary of factors that previous research has associated with teacher retention decisions. Following Cochrane guidelines, a systematic review of research reviews was conducted to accomplish this first goal. Based on a critical analysis of theories used to guide teacher retention research, the second objective was to explore the potential of Social Cognitive Career Theory (SCCT) to better explain how and why teachers decide to stay in or leave their schools. The third objective was to interpret the retention factors identified by the review of reviews through the lens of an education-specific adaption of SCCT we label Ed-SCCT. Results demonstrate that a large proportion of the most consequential factors associated with teacher retention are found within the school organization. Furthermore, Ed-SCCT demonstrated that retention factors, as they manifest themselves in schools, form a complex web of distal to proximal relationships potentially influenced by those in school leadership roles. Teacher retention policies and practice are likely to be much more successful when they acknowledge the complex interactive nature of most retention factors, as well as the central role of the school organization and its leadership in influencing how these factors connect to impact teacher retention.

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KEYWORDS

career decision theory, school leadership, systematic literature review, teacher retention

Context and implications

Rationale for this study: This review of reviews addresses a global policy challenge of teacher retention. Through synthesizing the results of research about factors influencing teachers' retention decisions, the review aims to better understand how those factors associated with the school organization and its members interact to shape such decisions.

Why the new findings matter: New findings indicate the significance of school organizations in retaining teachers and demonstrate the value of using social theory to explain how different retention factors and conditions interact to influence teachers' decision to move schools or leave teaching.

Implications for researchers, policymakers and practitioners: The current state of teacher retention research is data rich but theoretically underdeveloped. Our review identifies four features of the evidence base that undermine the likelihood of its value to educational practitioners and policymakers and provide recommendations for future research: (1) direct empirical testing of social theory to explain the dynamic complexity underpinning retention; (2) addressing the missing evidence about the role of teachers' personal goals in their retention decisions; (3) reducing variation in conceptions, methodology and outcomes that currently contribute to the challenge of interpreting results; and (4) unpacking what effective school leaders actually do to retain teachers.

Retaining skilled teachers in schools and in the profession is a longstanding and widespread challenge across school systems varying widely in income levels (e.g. UNESCO & International Task Force on Teachers for Education 2030, 2024). The challenge is especially significant among teachers in the first 5 years of teaching (e.g. Ingersoll, 2003; Schleicher, 2011) and among teachers of core subject areas such as maths, modern foreign languages and science (European Commission, 2012; European Commission, Directorate-General for Education, Youth, Sport and Culture et al., 2013). Schools in remote rural communities and urban schools serving socio-economically deprived communities are especially prone to unusual amounts of teacher attrition (e.g. Allen & McInerney, 2019; Boyd et al., 2008).

Excessive teacher attrition has negative effects on student learning, the working conditions of those teachers who stay in their schools, and the chances of allocating typically stretched education finances most productively. Using a distinction made by Ronfeldt et al. (2013), these negative effects are both *compositional* and *disruptive* in nature. *Compositional effects* are changes in the level of teachers' expertise in schools. If teachers leaving a school are less effective than those who replace them, there is a potential positive effect on students and perhaps the teachers who stay. However, as evidence provided by Ronfeldt et al. (2013) demonstrates, 'turnover has a harmful effect on student achievement, even after controlling for different indicators of teacher quality, especially in lower-performing schools' (p. 29).

Disruptive effects, felt mostly by teachers who stay in their schools, include increased difficulty establishing productive working relationships among their colleagues and between

staff and students. Disruptive effects also include reduced chances of developing high levels of reciprocal trust among staff, students and parents and increased difficulty in maintaining a focus on key school goals and priorities. Excessive levels of teacher turnover in schools are also financially disruptive. One US analysis, for example, estimated that 'teacher replacement costs, including expenses related to separation, recruitment, hiring, and training, can range from an average of \$9,000 per teacher in rural districts to more than \$20,000 in urban districts' (Darling-Hammond et al., 2017).

To help better understand the teacher retention problem and contribute solutions, this study addressed three research questions:

1. What factors have previous research associated with teacher retention decisions? Following Cochrane guidelines, a review of research reviews was conducted to address this first question.
2. How well do the theories used to guide the teacher retention research, evident in the review of reviews, explain how and why teachers decide to stay in or leave their schools? A critical analysis of those theories was undertaken to answer this question.
3. To what extent, and in what ways, does Social Cognitive Career Theory (SCCT) have more explanatory power than other theories used to guide teacher retention research? To answer this question, we first describe the broad outlines of SCCT as it has been applied to the career choices of employees in many organisational sectors. We then develop an education-specific adaptation of the theory, Ed-SCCT, and interpret the retention factors identified by the review of reviews through its lens.

SOCIAL COGNITIVE CAREER THEORY (SCCT)

Our search for a theoretical framework to explain how and why organizational members make their retention decisions resulted in the choice of Social Cognitive Career theory, especially the work of Lent and his colleagues (e.g. Lent et al., 2000, 2002). Exemplifying many of the features of positive psychology (Seligman, 2002), SCCT is rooted in social constructivist assumptions encompassed in general social cognitive theory (e.g. Bandura, 1986) about the agency of individuals in shaping their own behaviour. According to such theory, a considerable proportion of human behaviour can be explained by an array of beliefs about oneself, influenced by the contexts in which one finds oneself. Lent et al. (2000, 2002) explain that SCCT emphasizes both the direct and vicarious learning experiences that shape people's (a) sense of self efficacy, (b) outcome expectations, and (c) personal goals. These three variables are the basic 'building blocks' of career development and represent 'key mechanisms by which people exercise personal agency' (2002, p. 262).

A hallmark of SCCT has been efforts to better understand *conditions*—both *objective* and *perceived* environmental conditions—that facilitate or limit the exercise of human agency in career development (Lent & Brown, 2019). Lent and his colleagues (2002) argue that SCCT's attention to personal perceptions of the environment does not minimize the substantial impact of objective features of the environment, but it does highlight 'how individuals make sense of, and respond to, what their environment provides' (Lent et al., 2000, p. 37). Such a focus on the interplay between individuals' exercising agency in their own career development and the environmental factors that promote or constrain agency is consistent with other theories and research about employee turnover and persistence (e.g. March & Simon, 1958; Peterson & Seligman, 2004).

Additionally, SCCT's theoretical roots in social *cognitive* rather than social *learning* theory have resulted in its strong conceptual foundation for better understanding the sociocultural, economic and policy conditions that shape the working conditions and learning opportunities

to which individuals are exposed as well as the specific cognitive, self-regulatory and motivational mediators through which work experiences influence their career behaviour and choices (Lent et al., 2002). Previous research on teachers' professional work, lives and retention (e.g. Day & Gu, 2010; Kazemi et al., 2022) concurs with SCCT's position (e.g. Lent & Brown, 2019, 2020) that coordinated differential work and learning experiences operate as intervening conditions that nurture or constrain individual teachers' self-efficacy (Zhang et al., 2024) and well-being (e.g. Hascher & Waber, 2021; Burns & Machin, 2013) in the school organization, and subsequently, influence their job satisfaction (Kasalak & Dağyar, 2020), an important predictor of retention (Geiger & Pivovarova, 2018).

Our review of reviews provided a lengthy list of factors (109) associated with teachers' retention decisions. However, none of these factors, alone or in combination, begins to capture the complex process that links even the most 'significant' (according to research) factors with a decision by teachers to stay or move schools or leave the teaching profession. A significant relationship between a teacher's perception of her/his workload as being excessive and a decision to move to another school, for example, is mediated by a 'black box' filled with other conditions which help explain the decision—*How much do I like my colleagues? How satisfied am I with my job aside from the workload?* and the like. Consideration of at least some of the conditions in this 'black box' will form part of almost all teachers' career decisions.

Following the review of reviews, the paper provides a critical analysis of theories identified in those reviews, describes our education-specific adaptation of SCCT (Ed-SCCT) which overcomes many of the limitations evident in those theories, and interprets the retention factors identified in the review of reviews through the lens of Ed-SCCT.

A REVIEW OF REVIEWS ABOUT FACTORS INFLUENCING TEACHER RETENTION

This review of research reviews followed the methodological guidance and expectations for Cochrane overviews of reviews (Pollock et al., 2023), using explicit and systematic methods to search for and identify published reviews related to teacher retention between 2013 and 2023.

Review methods

A comprehensive search strategy was used to locate published reviews in the multidisciplinary abstract and citation databases including Scopus, EBSCOhost (British Education Index, Educational Resources Information Centre), ProQuest (Education database, Social Science database) and Web of Science using the keywords: 'teacher retention' or 'teacher attrition' or 'teacher turnover' or 'teacher supply' or 'teacher shortage' or 'teacher leav*' or 'teacher stay*' or 'teacher mov*' or 'teacher mobility' or 'teacher quit*' or 'teacher dropout'. The search was further narrowed to papers identified as 'literature review' or 'research review' or 'systematic review' or 'meta-analysis'. Although this review focussed on teacher retention, keywords relating to teacher attrition and turnover were included to achieve a more comprehensive array of factors impacting teachers' decisions to move schools or leave the profession. The exact Boolean search string(s) used for each platform is presented in Table A1 in Appendix 1. For instance, in the Scopus database the search string included: TITLE-ABS-KEY ('teacher retention' OR 'teacher attrition' OR 'teacher turnover' OR 'teacher supply' OR 'teacher shortage' OR 'teacher leav*' OR 'teacher stay*' OR 'teacher mov*' OR 'teacher mobility' OR 'teacher quit*' OR 'teacher dropout') AND TITLE-ABS-KEY

((literature AND review) OR (research AND review) OR (systematic AND review) OR (meta-analysis)). The following inclusion and exclusion criteria were then applied to screen the reviews retrieved from our initial search (Table 1).

Document title, abstract and keywords searches were performed within citation databases in August 2023. For the years 2013–2022, the default publication year was selected in the aforementioned databases, including the entire year from 1 January to 31 December. For the year 2023, our search covered the period ranging from 1 January 2023 to 31 August 2023 when we began our systematic literature searches. These were followed by manual searches of peer-reviewed journals specializing in systematic reviews, that is, Educational Research Review, Review of Educational Research, Educational Review, Oxford Review of Education, and Review of Education, as well as the Cochrane Database of Systematic Reviews. The final step of the literature search involved hand searches of identified reviews' reference lists to ensure that additional reviews potentially eligible for this evidence synthesis were included.

Finally, a two-step screening process was conducted to identify reviews relevant to our research questions. Figure 1 presents the flow diagram of our review following the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines (Page et al., 2021). After removal of duplicates, papers were screened by the third author in two steps: (1) based on the title and abstract and (2) based on the full text. A sample of full papers was independently reviewed by the other two authors to reach decisions for inclusion and exclusion. This screening process resulted in the selection of 17 reviews for further analysis.

The digital tool Microsoft Excel was used to record the decisions made for each stage of screening and data coding. Table A2 in Appendix 1 describes the coding tool used for data extraction and Table A3 in Appendix 1 presents type of destinations, population of interest, review aim(s), review period, review sample size, sources of evidence, and types of evidence for all 17 reviews. Five types of literature reviews were included in our final sample of reviews: systematic reviews, meta-analyses, scoping reviews, critical reviews, and qualitative meta-analyses. Panel A of the table in Table A3 in Appendix 1 describes the three meta-analytic reviews and Panel B shows the six additional reviews that also conducted quality assessments of the original studies included in their reviews. Panel C describes the remaining eight studies that did not conduct such assessments and which played no further role in our analysis. As Table A3 in Appendix 1 indicates, the time period covered by studies

TABLE 1 Inclusion and exclusion criteria.

Inclusion criteria	Exclusion criteria
Published in English language between 2013 and 2023 (although the primary evidence included the selected reviews goes back to 1980s)	Published in language other than English and outside the timeframe of 2013–2023
Peer-reviewed journal article	Books, book chapters, grey literature
Review of empirical studies, including quantitative, qualitative and mixed methods research	Review of non-empirical studies, single studies that are not synthesis of evidence
Teachers in schools as a population of interest	Non-teachers
Focus on factors influencing teacher retention (staying in the same school or moving to a different school) and/or attrition (leaving teaching)	No clearly defined focus on teacher retention and/or attrition
A clear description of a search strategy and inclusion/exclusion criteria, so that we were able to map the nature and breadth of the evidence included in the reviews and assess the reviews for relevance and quality	No clear description of a search strategy and inclusion/exclusion criteria

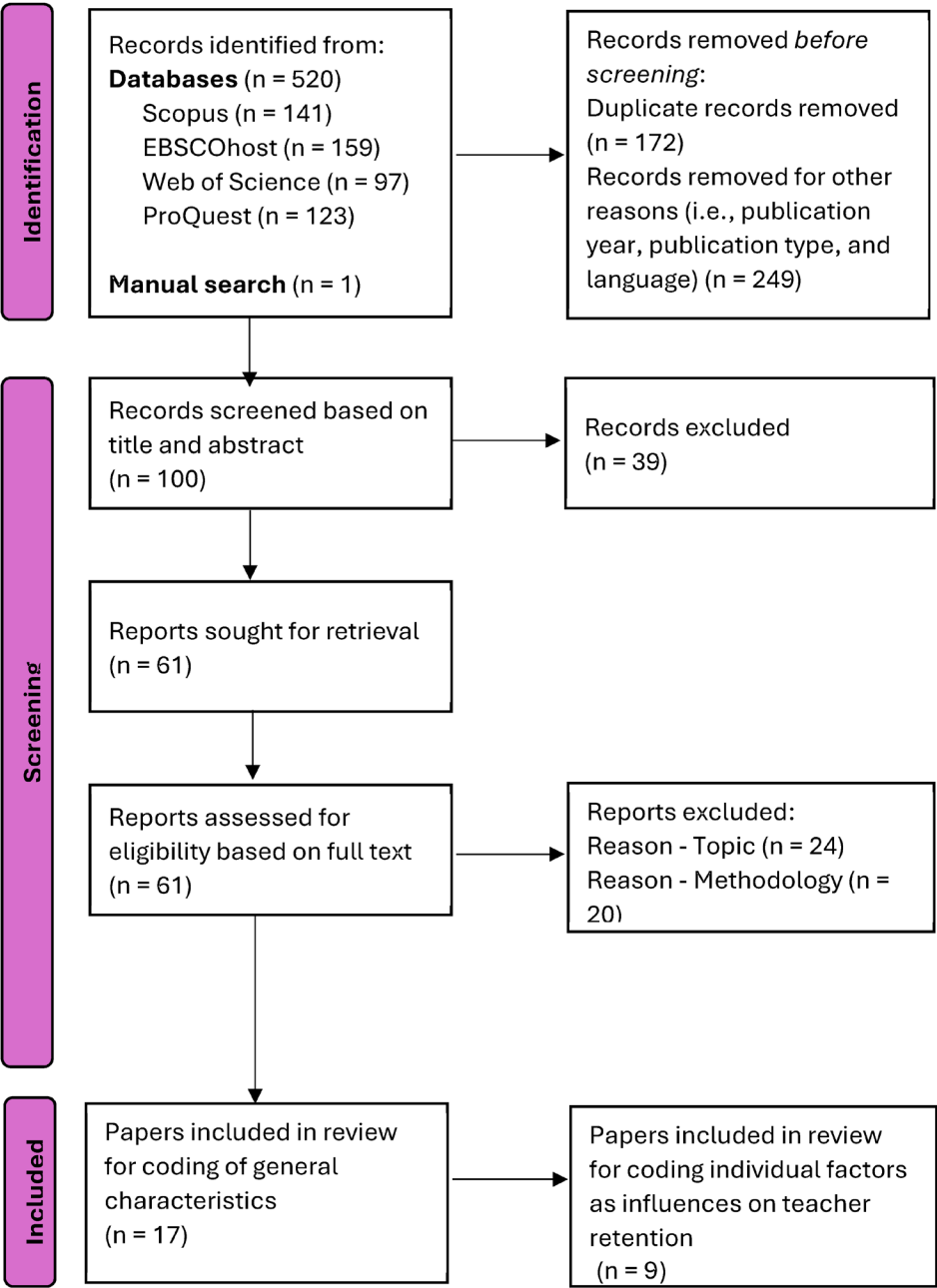


FIGURE 1 PRISMA diagram of review process.

included in the reviews ranged from 1980 to 2022, and the number of primary studies contained in each review ranged from 10 to more than 200. School teachers were the target population of all reviews and some focussed on specific groups of teachers, for example, special educators (Billingsley & Bettini, 2019), career-changers (Hogg et al., 2023), and teachers in rural schools (Rhinesmith et al., 2023). Eight reviews included quantitative,

qualitative and mixed methods evidence, five included only quantitative evidence, and four did not specify the type of research evidence reviewed.

Although the 17 reviews examined factors contributing to different retention choices, the distinction was often not acknowledged. Nine reviews focussed on teachers leaving the profession and four were concerned with both teachers moving schools (movers) and leaving the profession (leavers). Of these four reviews, only one made a clear distinction between movers and leavers in the presentation of the results. Finally, one review focussed on teachers staying in 'hard-to-staff' schools, while the remaining three reviews did not define the types of 'leavings' addressed in the studies they reviewed.

Quality assessment of the reviews

We performed quality assessment to evaluate the extent to which authors of the selected 17 reviews had conducted and reported their reviews with rigour and integrity. The assessment criteria drawn from the AMSTAR tool (Shea et al., 2009) and other systematic overviews of reviews (e.g. Smith et al., 2011) were used for the critical appraisal, focussing especially on the extent to which the primary research evidence included in each of the 17 reviews was sufficiently robust to support their conclusions. These quality assessment criteria included: (1) the extent of the search process reported in the reviews; (2) description of review selection and inclusion criteria; (3) appraisal of the methodological quality and risk of bias in the design, conduct, analysis and/or reporting of primary studies; and (4) evaluation of publication bias (Smith et al., 2011; see also Pollock et al., 2024). Application of these quality assessment criteria resulted in the selection of nine reviews for more detailed analysis, three of which are meta-analyses. The seven remaining reviews reported no assessment of the methodological quality and reporting rigour of the primary studies included in their analyses; results from these seven reviews are not included in subsequent reporting of results.

Analytic methods

Analysis of results reported in the nine reviews included coding of individual factors identified as influences on teacher retention and the classification of those factors into categories associated with the variables included in our Ed-SCCT (described later in the paper). In this process, our attention was placed upon examining the extent to which the identified teacher retention factors reflected those that Ed-SCCT considers central in shaping individuals' retention choices and exploring how to better understand the interactions between such factors in accounting for why teachers stay in or leave the profession. This process was organized in three steps:

Step 1: Coding of individual factors identified as influences on teacher retention and attrition.

Step 2: Grouping of identified factors into thematic categories outlined in Ed-SCCT.

Step 3: Mapping of the thematic categories against Ed-SCCT.

Focussed only on the nine reviews which included quality assessments, Table 2 lists the retention factors resulting from the reviews (left column) classified by the broad variables associated with Ed-SCCT and identified by at least four of the nine reviews or at least one of the meta-analytic reviews. This table also identifies factors explicitly influencing teachers who moved schools (movers) from factors influencing teachers who left the profession (leavers). One of these reviews identifies, in addition, factors that persuade teachers to remain in 'hard to serve' schools (See, Morris, Gorard, & El Soufi, 2020). Moreover, using Cohen's cut-off values (Cohen, 1988, 1992), Table 2 also reports the effect sizes of those factors

TABLE 2 Retention factors identified by at least four reviews or at least one of the meta-analytic reviews (effect sizes presented as S, small; M, medium; L, large).

Meta-analytic reviews		Non-meta-analytic reviews							
		'Leavers' (leaving the profession)		'Movers'	Leavers* (moving schools or leaving the profession)	Leaving/staying in the profession	Staying in 'hard-to-staff' schools		
Conceptual categories	Factors	Li and Yao (2022)	Madigan and Kim (2021)	Nguyen et al. (2020)	Nguyen et al. (2020)	Billingsley and Bettini (2019)	Palma-Vasquez et al. (2022)	Hogg et al. (2023)	See, Morris, Gorard, & Kokotsaki, & Abdi (2020)
School leadership	School leadership/principal leadership/administrative support			S	S	x	x	x	
	Retention bonus/financial benefits			S		x		x	x
External contexts	Union			S	S				
	Non-teacher salary			S					
School contexts	Teaching Salaries			S		x	x	x	
	School type (Charter v. traditional public; Private v. traditional public)			M			x		
	School level (High school v. elementary; Middle school v. elementary; Secondary v. elementary)			S			x		
	Urbanicity (Urban v. rural)				S	x			
	Poverty/socio-economic status				M	x	x		

TABLE 2 (Continued)

		Meta-analytic reviews	Non-meta-analytic reviews			
Teacher working conditions	Trust	L				
	School climate/work environment/school culture	M	S		X	
	Teacher collaboration/support among teachers/collegial support	M		X	X	
	Justice	M				X
	Late hiring					S
	Performance evaluation	S	S			
	Workload/workload manageability	S		X	X	
	Professional development					S
	Classroom autonomy/teacher autonomy			X	X	S
	Induction/mentoring		S		X	X
	Teaching materials/material resources		S		X	X
	Classroom assistant		S			
	Expenditure support		S			
Teacher self-efficacy	Principal effectiveness		S			
	Teacher effectiveness		S		X	
	Student behaviour/disruptive behaviour					
	Self-efficacy/teaching efficacy	S		X	X	

(Continues)

TABLE 2 (Continued)

		Meta-analytic reviews		Non-meta-analytic reviews	
Teacher well-being	Professional status	L		X	X
	Perceived stress	M		X	X
	Burnout	M		X	
	Emotional exhaustion/exhaustion	M		X	
	Depersonalization	M		X	
Teacher's motivational states	Pay satisfaction	S			
	Reduced accomplishment	S			
	Intrinsic motivation	M		X	
Teacher's personal and professional characteristics	Teaching experience	S		X	
	Gender	S		X	
	Age (continuous; greater than 28)	S		X	
	Race/ethnicity (Hispanic)	S		X	
	Race/ethnicity (Non-white)			X	
	Number of children	S			
	Young child	S			
	Marital status				
	Full time teaching	S			
	Graduate degree (MA/PhD v. none)				
	National board				

TABLE 2 (Continued)

		Meta-analytic reviews		Non-meta-analytic reviews	
Job satisfaction	Certification		S	X	X
	Internship		S		
	Ability (test scores)		S		
	Highly qualified		S		
	Specialty (STEM)		S	S	X
	Specialty (special education; other)		S		
	Job satisfaction/career satisfaction	M	M		X
	Commitment		M		X
	Work engagement		M		

Based on how 'leavers' are defined in these reviews, that is, teachers moving to a different school or leaving teaching. Only statistically significant meta-analytic results are reported in the table.

identified in the three meta-analytic reviews as S (small: 0.1–0.3), M (medium: 0.3–0.5) or L (large: 0.5 or greater).

This review was not registered. A PRISMA 2020 checklist (Page et al., 2021) can be found in Table A4 in Appendix 1.

RESULTS

Theories used in teacher retention research

The term ‘theory’ is used here, broadly and loosely, in reference to any organization of ideas ranging from a classification (or categorisation) system (a type of ‘weaker’ theory), a set of concepts and principles, a framework of related observables, to a set of plausible connections among variables encompassed in an empirically developed, mathematical formula to explain a specified social phenomenon (one type of ‘stronger’ theory). A significant proportion of teacher retention research identified in our review of reviews falls at the weaker end of this range. This is not to say it is without value; some of it is methodologically quite sophisticated and serves such practical purposes as informing relevant policies as, for example, See, Morris, Gorard, Kokotsaki, & Abdi’s (2020) analysis of the reliability of evidence and robustness of research designs used in 20 studies evaluating policies aimed at attracting and retaining teachers in hard-to-staff schools. For reasons explained below, however, greater attention to the use of stronger theory would help significantly advance our understanding of teachers’ retention decisions. The current state of teacher retention theory can be assessed as a function of (a) the *extent* to which teacher retention research is theory informed, (b) the *nature* of the theories doing the informing, and (c) the *purposes* served by theories.

On the extent to which teacher retention research is theory informed, among the nine high-quality reviews of literature included in our review of reviews (below), four mentioned theories used in papers included in their reviews (Billingsley & Bettini, 2019; Li & Yao, 2022; Madigan & Kim, 2021; Nguyen et al., 2020). Billingsley and Bettini (2019) were the most explicit, noting that of the 30 studies in their review, only 10 made explicit use of theory. No mention was made of retention theory in the texts of the reviews by Unda (2023), See, Morris, Gorard, Kokotsaki, & Abdi (2020) and Palma-Vasquez et al. (2022). Hogg et al. (2023) recommended more attention to theory in future research without otherwise dealing with it. In sum, the evidence we examined suggests relatively modest use of theory in empirical research on teacher retention.

On the nature of theories used in teacher retention research, Vagi and Pivovarova (2017) concluded from their review of 40 studies that the range of theories used was quite restricted. However, the 10 theory-informed studies in Billingsley and Bettini’s review used a significant array of theories, for example, *conservation of resources* theory (Hobfoll, 2011), *affective events* theory (e.g. Junça-Silva et al., 2021), *ecological systems* theory (Burns et al., 2015), *human capital theory* (e.g. Gillies, 2015) and *sense-making theory* (e.g. Weber & Glynn, 2006) and *economic labour market theory* (Guarino et al., 2006).

These theories have been applied in many contexts and in response to many problems in addition to teacher retention. While this range of application can be viewed quite positively, theories of this type require significant adaptations to serve as guides to teacher retention research. We use *economic labour market theory* to illustrate this point. This theory suggests that the willingness of individuals to obtain the necessary qualifications and work as teachers depends on their rational assessment (based on radically incomplete information) of the desirability of the teaching profession relative to alternative opportunities (Guarino et al., 2006, p. 177).

This theory has at least three limitations when it is applied to teacher retention. First, it ignores well-known intrinsic motivations for entering the profession such as admiration for a teacher of one's own, desire to work with children, altruism, and following in the footsteps of a family member (e.g. Domjan, 2018). Second, the theory also fails to acknowledge perspectives on teaching as a vocation as distinct from a job, occupation or profession and the differences such perspectives have on motivations to enter, stay or leave teaching (Hansen, 2021). Finally, while the theory may offer some useful insights into patterns of teacher recruitment, it is much less useful in relation to teacher retention. Using evidence from social and economic research, Song et al. (2020) demonstrate that once teachers' basic needs (as measured by relative incomes) have been satisfied, differences in job satisfaction, a strong predictor of retention, are more likely to be explained by variables closely related to the vocational nature of teaching, than other economic factors. As this account of economic labour market theory illustrates, many theories used to explain teacher retention require considerable adaptation if they are to be 'fit for purpose': they should explicitly acknowledge key features of the contexts in which teachers work and carefully consider the meaning and characteristics of the teaching profession.

Theories can serve descriptive, predictive and explanatory purposes (Biesta et al., 2011). Theories included in our review of reviews, for example, included those serving both descriptive and predictive purposes. Theories serving primarily descriptive purposes consist of logical or grounded categories of retention factors (e.g. Hogg et al., 2023). Theories of this type help researchers organize variables of interest in their studies. When relationships among those categories of variable are identified, the theories begin to serve predictive purposes. Ingersoll et al. (2019) provides a sophisticated example of a predictive theory. So too do the three meta-analytic reviews of Nguyen et al. (2020), See, Morris, Gorard, Kokotsaki, & Abdi (2020), and Li and Yao (2022) analysed in the next section of this paper; these meta-analyses test hypothesized relationships among categories of variables and assess their effects on teacher retention.

Explanatory theories that acknowledge the contexts in which teachers work are rarely found in teacher retention research. While predictive theories have significant practical value, explanatory theories offer the possibility of improving prediction by offering accounts of *how* and *why* those variables and relationships influence the desired outcome. As researchers engage in theorizing, they broaden and deepen how they interpret and make sense of the specified social phenomenon in situ (Biesta et al., 2011). One of our core assumptions, therefore, is that further progress in understanding teachers' retention decisions depends on the use of 'fit for purpose' explanatory theories. Theories of this type will (a) acknowledge those factors currently known to have the greatest influence on retention decisions, (b) specify the often-complex relationships and connections among those factors, and (c) provide explanations for why those relationships exist in teaching. Those theories should also be empirically testable.

One of the most important features of the modified SCCT theory used in this paper to understand how teacher retention decisions are made is its reflection of evidence about both the processes and variables associated with teachers' career decision-making, *in general*, and evidence from the review of reviews about factors identified as most influential in teachers' retention decisions, *in particular*. Combining these two sources of evidence has two advantages: it provides more coherent explanations for the nature and size of the effects of factors identified by retention research, and it has the potential to identify other important factors that retention research has yet to reveal.

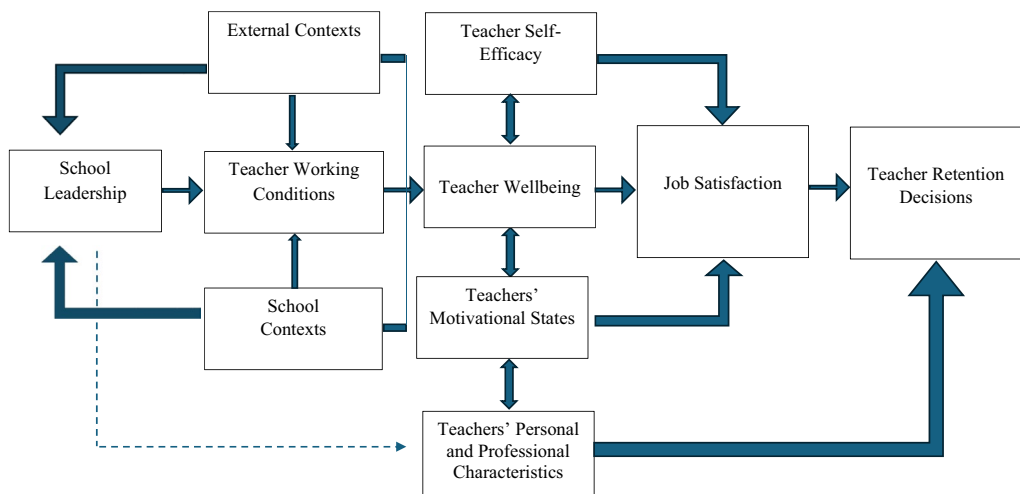


FIGURE 2 Explaining variation in teacher retention decisions.

ED-SCCT: AN EDUCATION SPECIFIC ADAPTATION OF SCCT

Figure 2 summarizes the variables and relationships associated with Ed-SCCT. This adaptation of SCCT for use in school contexts builds on the fundamental premises of SCCT but interprets several of its variables differently (*Teachers' Motivational States*), whereas several others have no direct parallel in SCCT (*Teacher Well-Being* and *Job Satisfaction*). Reflecting a considerable amount of school-specific evidence in educational research (e.g. Day et al., 2016, 2024; Grissom et al., 2021; Leithwood & Jantzi, 2006), our adapted Ed-SCCT considers social cognitive assumptions about the roots of human agency and privileges proximal influences located in the school organization over more distal influences on teachers' retention decisions. Ed-SCCT also recognizes that *School Leadership* is positioned to play an, as yet, underappreciated role in accounting for teacher retention and awards an outsized role for school leadership in either directly or indirectly shaping most other influences on teacher retention.

Ed-SCCT adds an *explanatory* dimension to other conceptual frameworks used in the reviews (Li & Yao, 2022; Nguyen & Springer, 2023) that are primarily descriptive, although in some cases also predictive in nature. For example, Nguyen and Springer (2023) describe the evolution of conceptual frameworks about teacher retention since 1958 through six stages and three broad phases. This account identifies increasing numbers of factors influencing teacher retention, each rooted in a theory justifying its consideration for retention research. But the outcome of this evolution remains a list of factors classified by the theories which identified them, not an explanation of how teacher retention decisions are made through the *interaction* of these factors.

Rather than fixating on an exhaustive array of discrete factors, Ed-SCCT provides a solution by creating a coherent explanatory system to address the interactions between them. Its theoretical framing is grounded in compelling research evidence about the significant influence on teacher retention decisions of conditions found in teachers' immediate work environments (e.g. Borman & Dowling, 2008; See, Morris, Gorard, Kokotsaki, & Abdi, 2020) and acknowledges consistent findings about the importance of 'administrative support' (e.g. Grissom et al., 2021) while clarifying what such support entails. It also privileges those cognitive and affective states of teachers directly influencing their retention decisions such as job satisfaction (e.g. Madigan & Kim, 2021), self-efficacy (e.g. Zhang et al., 2024), and well-being at work (e.g. Song et al., 2020). The conceptual order imposed by Ed-SCCT on these

variables helps to reveal that some seemingly direct influences of teacher retention (e.g. job satisfaction) are themselves outcomes of some variables and processes that may have much more far-reaching and deep influence on the whole systems of change.

USING ED-SCCT TO EXPLAIN TEACHER RETENTION DECISIONS

For each of these variables identified in [Figure 2](#), this section clarifies the meaning of the variable, cites evidence justifying its relationship with those other variables to which it is linked in [Figure 2](#), and highlights the specific retention factors associated with the variable, identified by at least four of the nine reviews or at least one of the meta-analytic reviews ([Table 2](#)).

Conditions in the school

The left side of [Figure 2](#) indicates that, in addition to school leadership, there are three variables *in the school organization* and its external contexts that influence one another and have indirect impacts on teacher retention through a handful of variables *in the teacher*. School leaders have an important role to play in shaping teachers' perceptions of contextual variables both internal and external to their schools. Such perceptions help shape teachers' interpretations of their working conditions.

School leadership

Ed-SCCT defines leadership as the exercise of influence towards the achievement of organisational goals. Cancio et al.'s (2013) review concluded that lack of leadership support is one of the most frequently cited causes of attrition. Such lack of support entails school leaders having competing priorities, being insensitive to both the needs and expertise of teachers, and failure to provide teachers with meaningful feedback. In contrast, a sizeable corpus of evidence indicates that supportive school leadership has an outsized although largely indirect impact (Hallinger & Heck, 2011) on the retention of teachers and the success of students. For example, Boyd et al. (2011) found that:

'teachers' perceptions of the school administration had by far the greatest influence on teacher retention decisions. This effect of administration is consistent for first-year teachers and the full sample of teachers and is confirmed by a survey of teachers who have recently left teaching.'

(p. 303)

These leaders help to optimize the status of a large handful of working conditions that foster the retention of experienced and skilled teachers (e.g. Eleftheriadou et al., 2025; Grissom et al., 2019; Leithwood et al., 2017).

Teachers' perceptions, in this study, were concerned especially with the 'support' teachers received from their administrators. While taking many different forms, such support entails helping make teachers' working conditions easier and more effective, hence, the direct connection in [Figure 2](#) between school leadership and those working conditions. Many of the forms of support provided by successful school leaders are characteristic of *transformational* approaches to leadership. Highlighting the impact on teacher retention of such approaches to leadership, Ladd (2011) notes such practices as promoting collaboration and trust among

teachers, engaging the school community in building shared visions and fostering teacher autonomy in classroom decision-making and participation in school-wide decisions.

School leadership was identified as an influential factor by five of the study's nine reviews. Palma-Vasquez et al. (2022) report that lack of support from leadership, administration or management is a determining factor in teachers leaving the profession and moving schools. In their meta-analytical study, Nguyen et al. (2020) found that the odds of teachers leaving schools (leavers and movers) with stronger administrative support are 0.80 times the odds of teachers leaving schools (leavers and movers) with weaker administrative support.

Results related to specific groups of teachers also highlight the significance of school leadership in teachers' 'leaving' decisions. Specifically, studies examining global ratings of school-based administrative support found that special educators were more likely to stay when they rated administrative support more highly (Billingsley & Bettini, 2019). According to qualitative research, the lack of administrative support is a significant school-level challenge for Latinx teachers leaving the profession (Unda, 2023). Finally, the perceived quality of school leadership is a major source of dissatisfaction for some career-change teachers (Hogg et al., 2023). Other features of school leadership with negative influences on teacher retention decisions identified by fewer than four reviews included late hiring practices, problematic performance evaluation practices, and the perceived unfairness of decision-making processes.

External contexts

Two variables in Figure 2 replace SCCT's 'Background Contextual Affordances' with affordances that are external to the school (External Contexts) and those that are specific to teachers' own schools (School Contexts). Figure 2 indicates that school leaders play a key role in interpreting the significance, for their schools, of the External Contexts aimed at changing school practices (e.g. state accountability policies), as well as helping to shape and manage objective features of the school organization itself—the School Contexts (e.g. student intake, student discipline policies). Enacting such features of the External Context as policies and other initiatives from outside the school is shaped by school leaders, especially principals (e.g., Coburn, 2005; J. Sun et al., 2023). How these leaders make sense of what a particular policy or other external initiative means to their schools and then decide 'whether and how to ignore, adapt, or adopt' the initiative locally (Seashore Louis & Robinson, 2012; Spillane et al., 2002, p. 733) influences how those External Contexts are interpreted by their teachers and how likely are those external initiatives to disrupt, constrain or advance further improvement of the school (e.g. Gu, 2023; Leithwood, 2018; Sinden et al., 2004).

One element of the school's External Context, financial incentives such as retention bonuses, was identified by at least four of the reviews. However, no clear benefits of such incentives were found for teachers staying in the profession or for keeping teachers in hard-to-staff schools (See, Morris, Gorard, & El Soufi, 2020). One of the meta-analytic reviews identified the prevalence of teacher unions and both teacher and non-teacher salaries, both with small effect sizes on teacher retention decisions (Nguyen et al., 2020).

Lower salaries are found to be associated with higher rates of both leaving the profession and moving schools (Palma-Vasquez et al., 2022). Nguyen et al. (2020) report that an increase in salary slightly reduces the odds of teachers leaving the profession and moving schools. However, while 'an increase in salary slightly reduces the odds of teacher turnover even though the result is statistically significant, the point estimate is close to one' (p. 10). Largely in agreement, See, Morris, Gorard, & El Soufi (2020) results focussed on challenging schools, in particular, concluded that 'on its own monetary inducements are clearly not enough to keep teachers in challenging schools or in difficult areas' (p. 693).

School contexts

A reflection of SCCT's Contextual Influences Proximal to Choice Behaviour, School Contexts are limited in Ed-SCCT to the objective features of the school organization. Some features of both the school and external contexts will be perceived by teachers as more or less difficult to manage; after SCCT, some will be perceived as barriers to their work while others will be considered helpful (Lent et al., 2000) and such perceptions will have important consequences on teachers' views of their working conditions.

In our review, only one meta-analysis identified any features of School Contexts influencing teacher retention decisions. Nguyen et al. (2020) reported small to medium effect sizes on teacher retention decisions of school type (charter/private/public), school level (elementary, middle, high school), school location (urban/rural), and school community demographics (poverty). The combined evidence from the primary studies included in Nguyen et al. (2020) suggests that teacher retention is higher in traditional public elementary schools in rural locations serving the students of medium to higher income families.

Teacher working conditions

Working conditions are defined broadly in Ed-SCCT as 'the physical features of the workplace, the organizational structure, and the sociological, political, psychological, and educational features of the work environment' (Ladd, 2011, p. 237). This variable encompasses but is not limited to SCCT's Learning Experiences. According to our theory, teachers' subjective beliefs about their working conditions have direct effects on their sense of Self Efficacy, Well-Being and Motivational States (e.g. Bowling et al., 2015; Hughes, 2012) and, through those teacher states, on their retention decisions. At least four of the reviews identified working condition likely to increase retention including access to induction/mentoring opportunities (e.g. Nguyen et al., 2020) as well as supportive environments and colleagues (e.g. Li & Yao, 2022). Nguyen et al. (2020) found that teachers with access to good in-service professional development were 16% less likely to leave the profession and move schools than were those without such access. Other teacher working conditions positively influencing (small effect sizes) teacher retention identified by one or two of the meta-analyses include the nature and quality of teaching materials, availability of classroom assistants, adequacy of budgets and the perceived effectiveness of the principal (Nguyen et al., 2020). Both Palma-Vasquez et al. (2022) and Li and Yao (2022) found that teachers' perceptions of an excessive workload had a strong influence on teachers' decisions to both leave the profession and move schools. In addition, teachers were marginally more likely to leave the profession and move schools when they were in schools experiencing more student disciplinary problems and disruptive behaviour (e.g. Nguyen et al., 2020; Palma-Vasquez et al., 2022). Teachers with less autonomy were also more likely to move or leave according to the reviews.

Conditions in the teacher

As Figure 2 indicates, Ed-SCCT indicates that teachers' perceptions of their working conditions (e.g. helpful, unhelpful) have a bearing on their sense of efficacy, well-being in school and motivational states. These three internal states have a significant bearing on teachers' levels of job satisfaction which, in turn, directly influences teachers' retention decisions.

Teacher self-efficacy

Ed-SCCT defines Self-Efficacy as teachers' self-referent judgements about their own abilities to achieve what they expect of themselves and likely what the school has established as its goals and priorities (Kelley & Finnigan, 2003; Zee & Koomen, 2016). Self-efficacious teachers suffer less emotional exhaustion and burnout, have more confidence in mastering challenging tasks and experience higher levels of commitment, personal accomplishment and sense of responsibility for teaching (e.g. Klassen & Chiu, 2010; Pillay et al., 2005; Schwerdtfeger et al., 2008; Skaalvik & Skaalvik, 2010; Song et al., 2020; Zee & Koomen, 2016). Considerable amounts of recent evidence, as Figure 2 indicates, also confirm a significant relationship between Teacher Self-Efficacy and both Job Satisfaction (Kasalak & Dağyar, 2020) and Well-Being (Gu et al., 2025; Zee & Koomen, 2016).

While evidence unambiguously points to teacher Self-Efficacy being an important explanation for much of what teachers do and think, our review of reviews suggests that it has not received the attention it deserves in research on teacher retention decisions. Only three of our nine reviews, including Li and Yao's (2022) meta-analysis, identified Self-Efficacy as an influence (a significant but small effect size) on teacher retention decisions.

Teacher well-being

After distinctions identified among many different types of well-being definitions by Hascher and Waber (2021), Ed-SCCT defines teacher Well-Being as a work-related subjective and contextual variable. Consistent with the social cognitive foundations of our theory, this conception of subjective well-being is indicated by teachers' feelings of belonging and being able to 'be themselves', as well as believing they have colleagues who care about them and treat them with respect (e.g. Viac & Fraser, 2020).

Hascher and Waber's (2021) review concluded 'that the support of an individual person and the existence of a supportive work environment—represented through shared visions and values, school culture, organizational justice, or support for autonomy—are of considerable importance' (p. 13) to teachers' sense of subjective well-being. Trusting relationships with colleagues have also been identified as especially important for teachers' well-being in their schools (Tsuyuguchi, 2023).

A positive sense of well-being in the teaching profession is indicated by teachers believing that they are good at helping students learn, have accomplished a great deal as teachers, and feel like their teaching is effective and helpful. Teacher burnout, stress, exhaustion and feelings of depersonalization are negative expressions of well-being (Hascher & Waber, 2021). These expressions of poor well-being have significant negative effects on job satisfaction.

Four of our nine reviews identified teachers' feelings of poor well-being, as important influences on their retention decisions. Two closely related symptoms of poor well-being, that is, stress and burnout, were reported in Li and Yao's (2022) meta-analysis to have medium (negative) effect sizes on teachers' intention to leave the profession. Two other symptoms of poor well-being, depersonalization and exhaustion, also were reported to have medium negative effect sizes on teacher retention decisions by Madigan and Kim (2021).

Teachers' motivational states

SCCT distinguishes among a persons' outcome expectations, interests and goals and describes how they are related. Our adaptation of this theory, Ed-SCCT, conceptualizes those

three states as a single multidimensional latent variable labelled Teachers' Motivational States. This variable reflects both Goal (e.g. Locke & Latham, 2015) and Expectancy-oriented theories of motivation (e.g. Vroom et al., 2015). Personal goals are significantly influenced by a person's interests and serve as the major determinant of their job satisfaction and 'actions' (e.g. to stay or leave a school or the teaching profession). Goal-oriented theories of human motivation privilege the impact of a person's goals on their behaviour. Expectancy theories explain variation in persons' motivations as a function of their beliefs about the likelihood that their own efforts will produce desirable outcomes.

A crucial purpose of leadership is to help create a compelling set of goals and overall sense of purpose that reflect both the needs of the organization and the interests and values of individual organizational members (J. Sun & Leithwood, 2015). Teachers' Job Satisfaction is significantly influenced by teachers' beliefs about the extent to which work in their schools and participation in the teaching profession provide opportunities to accomplish their personal/professional goals (e.g. Roberson, 1990).

As central to SCCT and our adaptation of it, only two of our nine reviews uncovered this variable in research about teacher retention decisions. For example, Li and Yao's (2022) meta-analysis reported that teachers' 'intrinsic motivation' had a medium effect size on teacher retention decisions.

Teachers' personal and professional characteristics

This construct reflects SCCT's 'Person Inputs' including, for example, teachers' years of experience, gender, ethnicity and personality (conscientiousness, see Bastian et al., 2017). Figure 2 indicates that personal inputs have a direct influence on both teachers' motivational states and their retention decisions (e.g. Bastian et al., 2017; Hughes, 2012). Evidence summarized in Table 2 associates 17 teachers' personal and professional characteristics with teacher retention decisions, all of which were reported to have at least a small effect size by the meta-analytic reviews. These characteristics vary greatly in terms of their susceptibility to influence from school leaders.

Figure 2 connects school leadership with Teacher Characteristics. Among those characteristics, it would be difficult in the short run, for school leaders to influence most of their teachers' age, gender, marital status, numbers of children or graduate status, although rapid teacher turnover would make it easier. The ethnic and racial makeup of school staffs is also malleable over time and a growing body of evidence suggests that teachers respond positively to working with ethnically/racially similar colleagues (e.g. M. Sun, 2018; Xu et al., 2024) although Nguyen et al. (2020) report that retention rates do not differ for Black vs. White teachers.

In the intermediate term, leaders are also able to influence at least some of their staffs' *professional characteristics*. Table 2 indicates that two meta-analytic reviews found that years of teaching experience (less likelihood of turnover over the age of 30) and standard teaching certification have positive effects on retention, while teaching specialties that might be useful in other types of organisations have negative effects by allowing teachers to pursue career options other than teaching.

Job satisfaction

Ed-SCCT defines Job Satisfaction as 'a positive or pleasant emotional state resulting from a person's appreciation of his/her own job or experience' (Demirtas, 2010). Longstanding evidence associates both Job Satisfaction (and Well-Being) with retention decisions (Butt & Lance, 2005; Doan et al., 2023; Struyven & Vanthournout, 2014). The literature on

both Job Satisfaction and Well-Being identifies two dimensions of each (Liu et al., 2020; Zakariya, 2020), one dimension related to membership in a school, and one dimension related to membership in the broader teaching profession. The same teacher might have high levels of Job Satisfaction and Well-Being on one dimension but not on the other.

Teachers' Job Satisfaction *in school* is the sense of fulfilment and gratification they experience through work in their schools. Although not included in Lent and colleagues' original version of SCCT, this form of Job Satisfaction includes feelings of enjoyment at work, the likelihood that teachers would recommend their schools to others as a good place to work, and teachers' unwillingness to change schools. Job satisfaction with teaching *as a profession* includes teachers believing the advantages of being a teacher clearly outweigh the disadvantages, agreement that they would choose teaching again if they had a chance to decide and not regretting their choice.

Job satisfaction was identified as an influence on teacher retention decisions by five of our nine reviews. Both Nguyen et al. (2020) and Madigan and Kim (2021) found that job satisfaction had a medium to small-sized effect on teachers' intentions to remain in the profession, an association attributed to enjoyment from their roles and the likelihood that they are more self-efficacious and enthusiastic. Similarly, Li and Yao (2022) found that job *dissatisfaction* was correlated with teachers' intention to leave the profession. These relationships seem not to be influenced much by the demographic and contextual moderators that have been examined in the literature to date (Madigan & Kim, 2021).

Teacher retention decisions

Evidence from the research reviews was concerned with two retention destinations in teachers' career decision-making: moving schools (*moving*) and leaving the teaching profession (*leaving*). While the consequences for schools of movers and leavers may be either positive or negative depending on their levels of expertise and the levels of expertise of those who take their place—Grissom et al.'s (2019) 'compositional' effect—the causes of each type of departure may be quite different, as would be the most likely initiatives aimed at addressing them (Gu et al., 2024). For example, unions, salaries, accountability policies, working hours and employment rates would seem to factor strongly into *leaving* decisions, whereas parent and community involvement, reduced paperwork, attractive teaching environments and promotion opportunities would seem to factor strongly into *moving* decisions.

That said, no clear evidence has emerged from our review of reviews about such differences between movers and leavers. For example, lack of teacher collaboration and collegial support were identified as contributing to both movers' and leavers' decisions in the three reviews of Billingsley and Bettini (2019), Nguyen et al. (2020) and Palma-Vasquez et al. (2022) but was reported by Hogg et al. (2023) to have influenced only leavers' decisions. Similarly, lack of professional development was found by Nguyen et al. (2020) to have a small effect size on intentions to move schools, but Billingsley and Bettini (2019) reported it as influencing the decisions of both movers and leavers.

DISCUSSION

This study aimed to synthesize the results of research about factors influencing teachers' retention decisions and to better understand, in particular, how and why those factors associated with the school organization and its members interact to shape such decisions. A 'review of reviews' methodology identified some 109 retention factors and an education-specific adaptation of Social Cognitive Career Theory, labelled Ed-SCCT, was used to

explain how school-related retention factors interact to shape teachers' retention decisions. Key to this explanation is the role that school leaders play in helping to shape teachers' perceptions about whether conditions in their schools, identified as 'retention factors' by the research, nurture conditions within teachers that directly determine their decisions to stay in or leave their schools or the teaching profession.

Factors with the greatest influence on teacher retention decisions

Most of the 109 retention factors were reported by only one or two reviews. Among the 12 most frequently identified factors in the nine reviews, as a whole, 11 were also reported to have at least statistically significant effect sizes on retention decisions by the three meta-analytic reviews (Table 2). Surprisingly, *Student Behaviour* did not rise to that level of influence, a finding at odds with evidence from the wider research literature reporting disruptive student behaviour as a major negative influence on teacher well-being and retention (e.g. Arthur & Bradley, 2023; Department for Education, 2018).

Seven of the 11 factors had only small effect sizes while *job satisfaction* and *stress* (a symptom of poor well-being) had medium effect sizes. Both the *professional status of teaching* and *trust in leadership* had large effect sizes according to the meta-analytic reviews. Interestingly, those two factors reported to have large effect sizes were identified by less than four of nine reviews. In the case of *trust in leaders*, only one meta-analytic review included that factor (Li & Yao, 2022) but its effect size on teacher retention decisions was large, reflecting broader evidence of its contribution to leader success (Sun et al., 2023).

These results indicate that the factors potentially influencing teacher retention decisions are numerous, and that some may yet to be discovered. Researchers' disciplinary expertise, theoretical interests and methodological preferences influence how and what they discover as the major factors of teacher retention. Results also indicate that the independent contribution of most of these factors to teacher retention decisions remains inconclusive. In our view, evidence from the nine reviews highlighted in this study provides very little guidance to educational practitioners or policy makers attempting to increase teacher retention.

The potential value of theoretical guidance

This study also demonstrates how the theoretical guidance provided by our adaptation of SCCT, i.e., Ed-SCCT, adds another level of meaning to the results of the retention research summarized in the review of reviews. By conceptualizing teachers' retention decisions as the outcome of complex social cognitive processes, each retention factor can be understood as having a more or less direct role to play in shaping that decision. The role of most of these factors is not to influence retention decisions directly (although the design of much retention research makes that assumption); it is to influence the *status of conditions* which either promote or discourage teacher retention. So, for example, ensuring that mentoring opportunities are provided to new teachers (a working condition) is both a consequence of supportive school leadership and a contribution to—among other things—teacher self-efficacy and sense of well-being. Mentoring matters not because it has direct effects on teacher retention decisions but because it has direct effects on several important downstream factors closer to that ultimate decision. Similarly, monetary incentives (a working condition) alone are found to have limited, lasting effects on retention.

Ed-SCCT begins to explain how and why different factors and conditions *interact* to influence teachers' decision to move schools or leave teaching. SCCT's emphasis on clarifying

conceptual linkages and paths of influence between major retention factors prompted us to look beyond the isolated effects of individual factors and examine how they influence teacher decisions directly and indirectly, and why. So, for example, the large effect size of *trust in leadership* on retention would become more meaningful for educational practitioners if it was understood not as a 'given' favourable working condition for retention, but the consequence of effective leadership that creates and shapes teachers' motivational states which, in turn, more directly influence retention decisions. Results of this review, therefore, also point to potentially important but neglected areas in the existing literature and in policy considerations on teacher retention.

Conditions in the school

Interpreting the retention factors identified by the review of reviews through the lens of Ed-SCCT revealed the extent to which the school organization moderated and shaped teachers' retention decisions. More than a third of the 109 factors, with almost half of the factors from the meta-analytical reviews (including the two showing large effect sizes), were associated with four features of the school organization—teachers' working conditions, school leadership and both external and school contexts. This finding strongly supports the observation that schools are finders and keepers of teachers (Johnson, 2019, 2004).

School leadership

While the review of reviews pointed to leadership-related factors with significant downstream effects on teacher retention, it did not award leadership the level of influence ascribed to by Ed-SCCT. Considerable evidence not accounted for in the review of reviews (e.g. Boyd et al., 2011), however, points to this as an important oversight in mainstream teacher retention scholarship. This evidence also speaks to an ongoing debate in the educational leadership research community about the relative value of instructional vs. transformational approaches to school leadership (Bellibaş et al., 2021).

When student achievement is the dependent variable, instructional leadership practices typically demonstrate greater impact (e.g. Robinson et al., 2008). But when teacher retention is the dependent variable, transformational leadership practices appear to matter most (e.g., Ladd, 2011). Taken together, these results endorse the value of 'integrated' models of successful school leadership (e.g. Printy et al., 2009). Improving instruction is obviously central to school improvement efforts but it is teachers who deliver that instruction. Retaining those who are especially skilled at that task, by supporting their work and helping to make it easier and more effective, 'tilts' the composition of school staffs over time towards turnover effects with significant positive consequences for students.

External and school-level contexts

Enacting policies in schools is an organizational behaviour which can be significantly shaped by school leaders, especially effective principals who, among other things, work with their teachers to set directions for the school and to redesign the organization to help accomplish those directions. Effective school leaders play an integral role in mediating the external 'noises' of educational policies by creating innovative and collaborative social and intellectual environments that nurture, challenge and inspire teachers to learn, to grow, and to stay. Those knowing how to mediate policy demands and take advantage of funding

opportunities to prioritise the transformation of the capacities, conditions and cultures in the organization—irrespective of their contexts—tend to retain their committed teachers and keep getting better (e.g. Bryk et al., 2010).

Among the most direct influences on teachers' work lives are the basic features of their schools' organization. Evidence from the review of reviews was mostly focussed on school governance (Charter vs. traditional), level (elementary vs. secondary), location (urban vs. rural), and student demographic features (poverty). While evidence does link these features to variation in teacher retention, their influence is relatively small. These variables are largely unmalleable in the intermediate term but readily accessible to researchers in many databases, which seems to be the most obvious reason for the extent to which they have been studied.

Teacher working conditions

The review of reviews identified a large handful of teachers' working conditions we consider to be self-evident *objective* features of well-functioning school organizations. They reflect, for example, some of the central features of both 'learning organizations' and 'enabling bureaucracies'. Learning organizations, as Stoll and Kools (2017) explain, emphasize 'the importance of individual, group and organisational learning with inquiry, problem solving and experimentation as key drivers of change and innovation' (p. 217). These features of a learning organization are reflected in such retention factors identified in the review of reviews as mentoring and induction opportunities for newer teachers, ongoing professional development opportunities for more experienced teachers, and collaborative cultures. Enabling Bureaucracies aim 'to empower employees to achieve goals while maintaining structure and accountability' (Sinden et al., 2004, p. 196). Teacher working conditions associated with teacher retention that reflect Enabling Bureaucracies include quality teaching materials, availability of classroom assistants, sufficient autonomy and flexibility for instructional decision-making.

While evidence from the review of reviews indicates that these objective features of a well-functioning school organization do influence the retention decisions of a large proportion of teachers, our theoretical roots in social cognitive theory remind us that it is actually teachers' perceptions of these variables—as either attractors or repellers—that promote or constrain their exercise of agency to achieve personal and professional goals and ultimately motivate them to move schools or leave teaching. At least some of the teacher working conditions associated with learning organizations and enabling bureaucracies will be perceived as repellers by some teachers. These teachers, for example, do not want more autonomy, preferring instead, to let others make decisions on their behalf; some teachers prefer the isolation of their own classrooms and are not attracted by the potential support (or 'intrusion') of their colleagues (e.g. Johnson, 2003). This is an indication of just how complex it is to craft policies aimed at improving teacher retention on a large scale and the size of the challenge facing leaders in individual schools to successfully implement those policies. But no other role seems better positioned to address them.

Conditions in the teacher

Efficacy, well-being and job satisfaction

As Ed-SCCT indicates (Figure 2), efficacy, well-being and job satisfaction influence one another, are nurtured by many of the same conditions in the school, and combine to have

large direct effects on teacher retention decisions. A compelling example of evidence supporting the placement of job satisfaction as having the most direct effect of these three conditions on teacher retention is provided by Boyd et al.'s (2011) large-scale study which found that, among a substantial list of possible reasons for leaving their schools, dissatisfaction with the job was the main reason cited by both former and current teachers, by a large margin.

The review of reviews includes most of those conditions in the teacher, as well as some of the conditions in the school that nurture positive states of those conditions in the teacher. Corroboration of these conditions along with several additions can be found in research focussed directly on those states, as distinct from primarily retention decisions. By way of summary, these additional conditions include work perceived to be meaningful, professional status such as being appreciated by colleagues, students, parents and supportive, effective school leadership as well as efficient management and access to adequate teaching resources. Conditions in the teacher are also nurtured by opportunities for professional development and collaboration with colleagues. Disruptive student behaviour and excessive workloads have negative effects on these teacher beliefs and dispositions (e.g. Hascher & Waber, 2021; Ortan et al., 2021). With the exception of teachers' perception of their work as *meaningful* and *efficient management*, all of these conditions can be found, at least in an approximate form, among the factors listed in Table 2.

Teachers' motivational states

Ed-SCCT conceptualizes teachers' motivational states as a latent variable including teachers' personal goals, interests and expectations. These dimensions have a significant influence on teachers' job satisfaction and retention decisions depending on the extent to which teachers expect the work they do in the school allows them to achieve their own personal/professional goals and to be a good match to their professional interests. Domjan's (2018) meta-analysis of factors motivating teachers to choose teaching and to remain in the profession sums up the 'intrinsic' factors found to have the greatest influence on such motivation as 'the desire to work with children, intellectual fulfillment, social contribution [often referred to as 'altruism'], the vocation to teach, passion and love for children'. 'Extrinsic' factors identified in Domjan's review included 'status, authority, working conditions, safety at work, the influence of one's family and friends, the flexibility of working hours and the lifestyle' (2018, p. 82).

By far the bulk of evidence about what motivates teachers to enter and stay in the profession—and likely in their schools—is intrinsic factors such as the belief that teaching is a 'calling' and the desire to make a social contribution. This does not mean that conditions in the school have no influence on teachers' motivational states, however. These states will be positively influenced by many of the practices associated with transformational approaches to leadership as, for example, awarding teachers reasonable amounts of autonomy, especially in their instruction, encouraging teachers to participate in wider school decision-making processes, authentically engaging teachers in establishing school goals, providing teachers with the kind of feedback about their work that enhances their expectations of success, and providing opportunities for teachers to engage with their colleagues (Slemp et al., 2020).

The importance of intrinsic factors helps to clarify the ambiguous results of research about the role of financial incentives in considerations about teacher retention. Such incentives are more likely to work when they can be introduced in ways that improve teachers' motivational states, efficacious beliefs and sense of well-being (e.g. Gu et al., 2018; Matthews et al., 2014).

LIMITATIONS

We acknowledge that this review did not include books, book chapters, commissioned research reports or other forms of published works. Inclusion of these publications would broaden the evidence base for the review. However, because they had not undergone the peer-review process which, despite its own limitations, is believed to provide a trusted form of scientific communication (Kelly et al., 2014), we chose to include articles published in peer-reviewed journals only. Additionally, our quality assessment process had only focused on the appraisal of the 17 reviews, rather than the methodological and conceptual quality of the primary studies selected by each of the 17 reviews. We acknowledge that the limited number of available reviews as a result of our quality assessment did not offer a wide range of evidence sources for analysis.

Moreover, our search targeted outputs in English language only and we are aware that additional reviews had been published since the completion of our review. Future work should consider these new studies and those published outside the dominant lingua franca to reduce linguistic and publication bias and increase our understandings of the factors that influence teacher retention in different contexts.

CONCLUSION

Reflecting on his own leadership journey in education, Sir John Dunford (2016) argued that recruitment would be easier if schools and the education system at large placed 'retention first' (2016, p. 129) in their endeavours to develop a capable workforce. Achieving this goal, however, requires a nuanced and robust body of evidence to inform policy and practice. This concluding section identifies four features of the evidence base included in our review of reviews which undermine the likelihood of its value to educational practitioners and policy makers and provide recommendations for future research that would significantly improve its value.

Directions for future research

Direct empirical testing of Ed-SCCT

None of the nine reviews providing most of the evidence for our review of reviews (including the two that had presented a framework of analysis) used theoretically grounded and empirically warranted conceptual framing to shed light on the nuances of teacher retention decisions. Our use of Ed-SCCT illustrates the promise of using a theory-informed approach to investigate the nuances of teacher retention decisions: it does this by highlighting linkages between categories of major influences, encapsulating the connections between processes and outcomes of change, and formulating a testable theory of action that can be used to improve retention over time.

Lent and his colleagues have provided considerable empirical support for SCCT in a wide range of contexts. The adaptation of SCCT for school settings used in this study has received no such testing, as yet: it remains a theory, albeit one justified by considerable existing evidence. Conducting a direct test of the theory would not be an especially challenging task. Using well-crafted working theories of action to test against evidence is, after all, not new to educational research (e.g. Bryk et al., 2023; Day et al., 2016; Leithwood et al., 2020).

Some of the resources used for future inquiries about teacher retention should be devoted to providing robust evidence supporting, challenging and refining Ed-SCCT.

Filling key gaps

Ed-SCCT points to a central role for *Teachers' Motivational States*, which reflects both goal- and expectancy-oriented theories of motivation. The social cognitive roots of SCCT define goals as 'the determination to engage in a particular activity or to effect a particular future outcome' (Bandura, 1986), which 'constitute a critical mechanism through which people exercise personal agency or self-empowerment' (Lent et al., 2002, p. 263). The significance of personal goals and the expectation that they are achievable, along with self-efficacious beliefs, agency and performance accomplishments in shaping individuals' career development and choices has received substantial empirical support since the inception of SCCT 30 years ago.

In SCCT, personal goals form an important intermediate link between interests and actions highlighting that, although environmental events and personal history shape people's behaviour, such behaviour is also 'motivated, in large part, by people's self-directed goals and by the other social cognitive factors with which goals interrelate' (Lent et al., 2002, p. 263). Longitudinal evidence shows that goals are reliably linked to persistence behaviour; they are indicators of objective career success. Helping people set challenging work-related goals may be a key avenue to their persistent and later success (Lent & Brown, 2019). Addressing the missing evidence about the role of teachers' personal goals in their retention decisions suggests an additional direction for future inquiry.

Recognizing variation in conceptions and methods

While 10 of the most frequently identified retention factors had at least small but significant effect sizes in relation to teacher retention, how those effect sizes were calculated, as well as how those factors were conceptualized and measured, varied across studies making direct comparison unreliable. For example, the majority of factors reported in Li and Yao's (2022) and Madigan and Kim's (2021) reviews had medium to large effect sizes on teacher retention, whereas almost all of these same factors reported in Nguyen et al.'s (2020) review showed small effect sizes. Variation in conceptions, methodology and outcomes contributes to the challenge of interpreting results. Future research should aim to reduce such variation.

Unpacking what supportive and effective school leaders actually do

Five of the nine reviews identified administrative or leadership support as an important retention factor and one of the meta-analyses reported a small effect size (Nguyen et al., 2020). Ed-SCCT conceptualizes school leadership as the primary upstream influence on teachers' eventual retention decisions and evidence in our review of reviews did identify a large cluster of working conditions potentially facilitated by school leaders. Nonetheless, 'leadership support' might look quite different for different teachers—for example, an aggressive approach to student discipline for some, the awarding of considerable autonomy in the classroom for others, and the nurturing of professional learning communities for yet others. Although we have argued for more attention to transformational approaches to leadership, the most direct implication for school leaders might be as simple as recommending they appreciate the nature of support needed by each of their teachers and do their best to provide it.

Implications for policy and practice

Adopting an Ed-SCCT perspective in the teacher retention challenge brings with it a small number of large implications for both policy and practice. For district, state and national policy makers, Ed-SCCT highlights the gulf between the highly subjective influences proximal to teachers' retention decisions (e.g. well-being, job satisfaction) and the distant, typically weak purchase on those subjective influences that agencies and actors outside the school organization are likely to have.

Implications for policy

This study suggests three ways in which those engaged in policy development could improve their purchase on teacher retention in three ways:

1. Engage groups of school leaders in crafting policy from the outset. Ed-SCCT places school leaders in the pivotal position of interpreting the meaning of externally initiated policy and determining the shape it will take in their school organizations. As evidence in this study suggests, differences in the retention challenge school leaders face depends, for example, on differences in school communities (urban vs. rural), school levels (elementary, middle and high school) and school demographics (language and cultures). Policy makers should seek advice from school leaders working in these contexts about retention policy directions and how best to implement them.
2. Direct most retention policies at improving conditions in the school that significantly affect teacher retention decisions rather than at teachers directly. While few teachers will object, for example, to modest increases in salary, their retention decisions are much more influenced by manageable workloads, opportunities for new learning, and collaboration with colleagues. Scarce resources in the school diminish teachers' access to each of these key influences on their retention decisions.
3. Keep the main goal the main goal. Improving retention is the main goal not implementing any individual policy with fidelity. Modifications and refinements should be viewed as evidence of intelligent policymaking not poor planning. Include, as part of the policy implementation process, formative assessment and feedback loops about policy effects. Modify policy processes in response.

Implications for practice

One of the main contributions of Ed-SCCT to addressing the teacher retention challenge is its reinterpretation of labels associated with school leadership (e.g. 'administrative support'). School leadership is not simply one of 109 'factors' as the review of reviews implies; it is a critical starting point in a complex chain of associations explaining how and why teachers decide to stay in or leave their schools. The implications of this for school leaders are as follows:

1. A large proportion of the pre-service and in-service development provided to school leaders has been, and remains, overwhelmingly focussed on the improvement of instruction as a means of enhancing the achievement of students. As we noted in the introduction to this paper, however (e.g. Ronfeldt et al., 2013), teacher departures typically trigger a host of compositional, disruptive and financial consequences

impacting teachers' work lives, organizational cultures and resources. These consequences are almost always harmful to student learning.

So, the main implication for practice of this study is to focus a significant proportion of the ongoing professional development of school leaders on how to retain their teachers, especially their most competent teachers. Ed-SCCT identifies those features of schools and teachers that should serve as a focus for such professional development.

2. While evidence of the type described in this paper can provide generally applicable guidelines for practice, there remains considerable variation among individual teachers in what shapes those dispositions most directly determining their retention decisions as, for example, their sense of efficacy, well-being and job satisfaction. School leaders should be especially attentive to these dispositions in their day-to-day work with individual staff members and take steps to remediate sources of problems before they escalate.

In sum, our view of the current state of teacher retention research is that it is data rich but theoretically underdeveloped. Addressing this shortcoming holds the promise of extending what we know about influences on teacher retention and providing a much more practical understanding of how schools themselves can address the issue.

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The authors declare no conflict of interest.

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No primary data were used in this research. We have reviewed published journal articles.

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The authors have nothing to report.

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No AI tools were used in this manuscript.

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APPENDIX 1

TABLE A1 Search strings and platforms.

Database	Search field	Search string
Scopus		(TITLE-ABS-KEY ('teacher retention' OR 'teacher attrition' OR 'teacher turnover' OR 'teacher supply' OR 'teacher shortage' OR 'teacher leav*' OR 'teacher stay*' OR 'teacher mov*' OR 'teacher mobility' OR 'teacher quit*' OR 'teacher dropout') AND TITLE-ABS-KEY ((literature AND review) OR (research AND review) OR (systematic AND review) OR (meta-analysis)))
EBSCOhost (ERIC, BEI)	Title	TI ('teacher retention' OR 'teacher attrition' OR 'teacher turnover' OR 'teacher supply' OR 'teacher shortage' OR 'teacher leav*' OR 'teacher stay*' OR 'teacher mov*' OR 'teacher mobility' OR 'teacher quit*' OR 'teacher dropout') AND TI ((literature review) OR (research review) OR (systematic review) OR (meta-analysis))
EBSCOhost (ERIC, BEI)	Abstract	AB ('teacher retention' OR 'teacher attrition' OR 'teacher turnover' OR 'teacher supply' OR 'teacher shortage' OR 'teacher leav*' OR 'teacher stay*' OR 'teacher mov*' OR 'teacher mobility' OR 'teacher quit*' OR 'teacher dropout') AND AB ((literature review) OR (research review) OR (systematic review) OR (meta-analysis))
Web of Science	Title, abstract, keywords	'teacher retention' OR 'teacher attrition' OR 'teacher turnover' OR 'teacher supply' OR 'teacher shortage' OR 'teacher leav*' OR 'teacher stay*' OR 'teacher mov*' OR 'teacher mobility' OR 'teacher quit*' OR 'teacher dropout' (Topic) AND ((literature AND review) OR (research AND review) OR (systematic AND review) OR (meta-analysis) (Topic)
ProQuest	Abstract	Abstract t('teacher retention' OR 'teacher attrition' OR 'teacher turnover' OR 'teacher supply' OR 'teacher shortage' OR 'teacher leav*' OR 'teacher stay*' OR 'teacher mov*' OR 'teacher mobility' OR 'teacher quit*' OR 'teacher dropout') AND abstract ((literature review) OR (research review) OR (systematic review) OR (meta-analysis))
ProQuest	Title	Title ('teacher retention' OR 'teacher attrition' OR 'teacher turnover' OR 'teacher supply' OR 'teacher shortage' OR 'teacher leav*' OR 'teacher stay*' OR 'teacher mov*' OR 'teacher mobility' OR 'teacher quit*' OR 'teacher dropout') AND title ((literature review) OR (research review) OR (systematic review) OR (meta-analysis))

TABLE A2 Coding tool for data extraction.

Criterion	Coding variable	Categories/options
Study design and method—focus and aims	Type of destinations	Leaving the profession; moving schools; staying in the profession; type of destination not specified
Study design and method—review	Aims/Objectives Type of review Time covered Sources of evidence Types of evidence	Qualitative, quantitative, mixed methods
Study sample	Teacher focus Sample size	

TABLE A3 Description of review papers (n = 17).

Authors	Type of destination	Type of review	Teacher focus	Aims/objectives	Time covered	Sample size	Sources of evidence	Types of evidence
Panel A: Meta-analytic reviews (included in coding)								
1. Li and Yao (2022)	Leaving the profession	Meta-analysis	In-service or practicing teachers	Relationship between diverse antecedents and teachers' turnover intention	1989–2020	94	Articles	Quantitative
2. Madigan and Kim (2021)	Leaving the profession	Meta-analysis	Teachers	Relationship between burnout, job satisfaction and teachers' intentions to quit	1985–2021	24	Journal article or thesis/dissertation	Quantitative
3. Nguyen et al. (2020)	Leaving the profession, moving schools	Meta-analysis	Teachers in primary or secondary schools	Factors associated with teacher turnover	1980–2018	120	Journal articles, dissertations, reports, working paper, conference presentation	Quantitative
Panel B: Reviews with quality assessment (included in coding)								
4. Billingsley and Bettini (2019)	Leaving the profession, moving schools	Systematic review	Special educators in US schools	Factors associated with special educator attrition and retention	2002–2017	30	Peer-reviewed studies and grey literature	Qualitative, quantitative and mixed methods
5. Hogg et al. (2023)	Leaving the profession	Systematic review	Career-change teachers	Factors that motivate career-change teachers to leave teaching	1990–2022	48	Empirical studies in peer-reviewed journals	Not specified
6. Palma-Vasquez et al. (2022)	Leaving the profession, moving schools	Scoping literature review	Teaching staff, regardless of their level of practice	How mobility is understood and measured and identify the results of the related factors	2008–2018	213	Peer-reviewed journal articles	Qualitative, quantitative and mixed methods

(Continues)

TABLE A 3 (Continued)

Authors	Type of destination	Type of review	Teacher focus	Aims/objectives	Time covered	Sample size	Sources of evidence	Types of evidence
7. See, Morris, Gorard, Kokotsaki, & Abdi (2020)	Staying in the profession	Systematic review	Teachers in state-funded schools from early years to post-secondary	Impact of strategies or policy initiatives to improve the recruitment and retention of teachers	Not reported	120	Empirical studies	Quantitative
8. See, Morris, Gorard, & El Soufi (2020)	Staying in the profession ('hard-to-staff' schools)	Systematic review	Teachers in state-funded/government schools	The most promising approaches to attracting and retaining teachers in hard-to-staff areas	(not specified)–2018	20	Empirical studies	Quantitative
9. Unda (2023)	Staying in the profession	Critical literature review	Latinx teachers in the United States	Factors influencing the recruitment and retention of Latinx teachers	1990–2022	50	Peer-reviewed journal articles, books, book chapters	Qualitative, quantitative and mixed methods

Panel C: Reviews (not included in coding)								
10. Booth et al. (2021)	Retention/type of destination not specified	Scoping literature review	Mid-career teachers	Definitions of mid-career teachers in England and their retention and development	5 years	Not specified	Peer-reviewed evidence and grey literature	Not specified
11. den Brok et al. (2017)	Leaving the profession	Literature review	Beginning teachers in the Netherlands	Factors that influence early career teacher attrition	Not reported	Not specified	Governmental reports, professional journals or conferences	Not specified
12. Mason (2017)	Retention/type of destination not specified	Qualitative meta-analysis	Foreign language teachers	Insights into how foreign language teacher attrition and retention has been studied globally	2000–2016	10	Articles	Qualitative, quantitative and mixed methods

TABLE A 3 (Continued)

Authors	Type of destination	Type of review	Teacher focus	Aims/objectives	Time covered	Sample size	Sources of evidence	Types of evidence
13. Nguyen and Springer (2023)	Leaving the profession	Systematic review	Teachers in K-12 education	Conceptual framework of teacher turnover	(40 years)–2018	159	Empirical studies	Qualitative, quantitative and mixed methods
14. Rhinesmith et al. (2023)	Leaving the profession, moving schools	Systematic review	Teachers in rural schools	Literature around teacher recruitment and retention in rural school settings	1980–2022	94	Articles	Qualitative, quantitative and mixed methods
15. Ruitenburg and Tigchelaar (2021)	Retention/type of destination not specified	Literature review	Certified second-career teachers	How certified second-career teachers' retention is influenced in their induction period in secondary education	2008–2018	27	Journal articles, doctoral dissertations, research reports	Qualitative, quantitative and mixed methods
16. Shockey et al. (2013)	Retention/type of destination not specified	Qualitative meta-analysis	Secondary school teachers	Relationships between teacher induction programmes, teacher motivation, job satisfaction and teacher retention rates	10 years	10	Government reports and white papers, peer-reviewed research articles and books	Qualitative, quantitative and mixed methods
17. Vitek (2015)	Leaving the profession; retention	Critical literature review	Special education teachers	variables that lead to special educator attrition and retention	2004–2015	34	Peer-reviewed journal articles	Not specified

TABLE A4 PRISMA 2020 checklist.

Section and topic	Item no.	Checklist item	Location where item is reported
Title			
Title	1	Identify the report as a systematic review	1
Abstract			
Abstract	2	See the PRISMA 2020 for Abstracts checklist	1
Introduction			
Rationale	3	Describe the rationale for the review in the context of existing knowledge	3–4
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses	4
Methods			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses	7
Information sources	6	Specify all databases, registers, websites, organizations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	6–7
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	6–7, Table A1 in Appendix 1
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	8
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	8
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	8
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	8 Table A2 in Appendix 1

TABLE A4 (Continued)

Section and topic	Item no.	Checklist item	Location where item is reported
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	9
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	9–10
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	9–10
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	10
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	8–10
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity and software package(s) used.	10
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	NA
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	NA
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	NA
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	9, Table 2
Results			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	Figure 1
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	10
Study characteristics	17	Cite each included study and present its characteristics.	Table A3 in Appendix 1

(Continues)

TABLE A4 (Continued)

Section and topic	Item no.	Checklist item	Location where item is reported
Risk of bias in studies	18	Present assessments of risk of bias for each included study	10
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots	Table 2
Results of syntheses	20a	For each synthesis, briefly summarize the characteristics and risk of bias among contributing studies	15–25
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect	NA
	20c	Present results of all investigations of possible causes of heterogeneity among study results	NA
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results	NA
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed	NA
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed	NA
Discussion			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence	25–32
	23b	Discuss any limitations of the evidence included in the review	33–35
	23c	Discuss any limitations of the review processes used	32
	23d	Discuss implications of the results for practice, policy and future research	35–37
Other Information			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	10
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared	NA
	24c	Describe and explain any amendments to information provided at registration or in the protocol	NA

TABLE A4 (Continued)

Section and topic	Item no.	Checklist item	Location where item is reported
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review	Funding statement
Competing interests	26	Declare any competing interests of review authors	Conflict of interest disclosure
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms, data extracted from included studies, data used for all analyses, analytic code, any other materials used in the review	Table 2 , Appendices 2, 3